Local–Global Feature-Aware Transformer Based Residual Network for Hyperspectral Image Denoising

Fengfeng Wang, Jie Li, *Member, IEEE*, Qiangqiang Yuan, *Member, IEEE*, and Liangpei Zhang, *Fellow, IEEE*

***Abstract*— Hyperspectral images (HSIs) are generally distorted by various types of damage and degradation due to limited imaging conditions. Hence, noise reduction is an essential process before HSI interpretations and applications. In this article, a novel local–global feature-aware transformer based residual network (FATR) is proposed for HSI denoising. First, a spatial–spectral feature extraction module is built to extract spatial and spectral shallow features simultaneously. Second, these spatial–spectral features are forwarded to the deep feature extraction module, which contains several local–global feature-aware transformer blocks, where contextual information as well as local and global information can be further aggregated by multiscale windows transformer layers. Finally, in the reconstruction module, different hierarchical features from branches of two modules are merged into the final restoration to recover clean HSIs. Extensive experiments on both synthetic and real-world data demonstrate that the model has a better ability to restore HSIs in terms of evaluation metrics and visual assessments.**

***Index Terms*— Hierarchical features, hyperspectral image (HSI) denoising, multiscale windows, self-attention, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) simultaneously record spatial images and spectral profiles with a high spectral resolution, which allows them to contain abundant information. Thus, HSIs can be used in numerous remote sensing applications, such as classification [1], [2], [3], unmixing [4], anomaly detection [5], and other applications [6], [7]. However, HSIs are inevitably subject to various noise damages and degradation due to limited imaging conditions, the mechanical limitations of hyperspectral imaging spectrometer, disturbances during data transmission, or photon effects [8], [9], [10], and these degraded HSIs can make subsequent applications problematic. At present, hardware-based noise suppression solutions are still costly and device-specific [8], [9], [10]. Therefore, noise reduction of HSIs is a critical process prior to HSI analysis and interpretation.

From a general perspective, two primary sources contribute to the noise in hyperspectral remote sensing images. One part of it comes from the observation environment. Issues such as natural lighting dramatic changes or insufficient and other problems during imaging will cause the introduction of random noise to varying degrees. Furthermore, the absorption of radiance at different wavelengths by the atmosphere or the surface in the natural environment tends to reduce the radiance received by the hyperspectral sensor and lower the signal-to-noise ratio (SNR). The other part comes from the imaging system itself. Inconsistent response of probe elements during the scanning and imaging process may occur, thus resulting in stripe noise [13] in the acquired images with light and dark distribution. Moreover, sensor temperature variations, periodic shifts, electromagnetic interference, short exposure times, and narrow bandwidths [14], [15] can also generate various kinds of noise, such as random noise, impulse noise, deadlines, and dead pixels.

The presence of different noise sources in an HSI makes its modeling and denoising task highly challenging [16]. Nevertheless, many methods have been proposed to solve this problem in recent decades [17], [18], [19], [20], [21], [22], [23], [24], [25]. Among these approaches, two general categories can be distinguished, viz., model-based approaches and deep learning-based approaches. In the model-based approach, the hyperspectral image denoising problem is generally regarded as an ill-posed inverse problem. By introducing appropriate prior constraints, such as nonlocal similarity [26], [27], total variation (TV) [19], [21], [28], and low-rankness [20], [25], the ill-posed problem can be transformed into a well-posed one. These prior terms can describe the local structure and detailed features of the image clearly. The model-based approaches generally have clear physical implications, but they usually involve time-consuming optimization processes and rely on manually designed complex priors. As an emerging paradigm in recent years, most deep learning-based approaches [8], [11] use a given noise-labeled HSI to make the obtained HSI from the network closer to the corresponding clean HSI of the corresponding target for parameters learning and training of

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denoising models. In contrast to regularization-based modeling methodologies, the structure of deep learning approaches is universal. It does not rely on physical degradation mechanisms, and denoising can be performed in an end-to-end manner when sufficient samples are provided. These deep learning methods usually contain a large number of parameters with uncertain meanings, and their models are highly complicated and poorly interpretable. Nevertheless, these methods have excellent results and extremely fast processing speeds.

Implementations in deep learning, especially in computer vision, convolutional neural networks (CNNs) have dominated since their development [29]. Moreover, they have been widely used in low-level vision and remote-sensing image processing [30], [31] in recent years. However, the evolution of network architectures has taken a different path in the field of natural language processing (NLP), where the leading structure is the transformer [32] network. The outstanding results of the transformer and the pure use of self-attention have generated a surge of research interest in the computer vision community to adapt these models to vision tasks [29], [33], [34]. Currently, transformer structures have been applied in many remote sensing image processing tasks, while transformer method for HSI denoising is still rare. In view of this current situation, we seek to extend the transformer backbone by extracting spatial and spectral features from the tensor data of HSI so that it can be used for HSI denoising.

In this article, we propose a local–global feature-aware transformer based residual network (FATR) for HSI denoising. Specifically, we first build a spatial–spectral feature extraction module to extract spatial and spectral shallow features. Thereafter, we construct a local–global feature-aware transformer based residual block (FATRB), which contains different types of window-based transformer layers. With these designs, useful contextual information as well as local and global information can be further excavated. Within these blocks, the spatial-lightweight operation is added in self-attention to reduce computational and memory costs, and residual connections are used to ensure training stability. Additionally, we put a convolution layer at the end of the block to augment the features. Then, by adopting several of these transformer blocks, a deep feature extraction module is, thus, formed to extract hierarchical features. Finally, in the reconstruction module, different features from both branches of the two modules are merged into the final restoration to produce a high-quality HSI reconstruction.

The main contributions of this work are summarized as follows.

1) As far as we know, this article is an earlier work employing the transformer structure with residual connections on HSI restoration. A spectral embedding operation is proposed to extract spectral–spatial features from HSIs, and a spatial-lightweight operation is used in self-attention to reduce computational and memory costs for HSIs.
2) In order to extract information from different regions of the feature map, we adopt a multiscale windows partitioning scheme, which improves the modeling and information extraction capabilities considerably.
3) The synthetic and real-world HSI experiments indicate that our proposed method can remove hybrid noise as well as severely Gaussian noise and maintain the relative restoration of the spectra without distortion compared with previous methods.

The remainder of this article is as follows. In Section II, the HSI degradation model, recent work on HSI denoising, and the development of the visual transformer will be introduced. In Section III, the method proposed in this article and related details will be presented. In Section IV, the experimental results on synthetic and real-world data are demonstrated. Finally, the conclusions will be given in Section V.

II. Related Work

The observed HSI is noted as $\mathcal{Y} \in \mathbb{R}^{H\times W\times B}$, where H and W denote the spatial size and B is the number of bands, representing the size of the spectral dimension. The noise degradation model of the HSI can be described as

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{X} \in \mathbb{R}^{H\times W\times B}$ is the ideal case of a noise-free HSI, and $\mathcal{N} \in \mathbb{R}^{H\times W\times B}$ is the additive noise, such as densely distributed Gaussian noise, sparsely distributed stripe noise, and dead pixels. In order to obtain noise-free data $\mathcal{X}$ from (1) with known $\mathcal{Y}$, many scholars have proposed various models for HSI denoising.

A. Model-Based HSI Denoising

Currently, numerous methods have been proposed for HSI denoising. As HSI can be treated as a stack of 2-D grayscale images, the most straightforward way of denoising is to use traditional single-band image denoising methods such as the nonlocal means (NLM) filter [35], the weighted nuclear norm minimization (WNNM) [36], and the block matching and 3-D filter (BM3D) [26] to remove noise from HSIs band by band. To make better use of the abundance of spectral bands, many improvements have been proposed, such as block matching and 4-D filtering (BM4D) [37], which considers similar 3-D image blocks to form 4-D data, and then co-filtering the grouped data based on a 4-D transform. To make full use of the prior information on HSI, several effective regularization methods have been used to process HSI, of which total variation and low-rank (LR) regularization are currently the most useful and robust methods. Total variation regularization is widely used for image denoising because of its ability to remove noise while maintaining the edge and detailed information of the image. As a result, many methods employ the total variation model to remove noise from HSIs, such as hyperspectral TV (HTV) [19] model. These total variation models can better preserve the segmented smoothing properties spatially. However, they ignore the essential spectral correlation properties in HSI. To overcome this drawback, spectral correlation is introduced in the design of total variations, such as the spectral–spatial adaptive hyperspectral TV (SSAHTV) [19] model, the anisotropic spectral–spatial TV (ASSTV) [28] model, and the enhanced 3-D TV (E3DTV) [38] model.

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Furthermore, the low-rank model is one of the widely used HSI denoising methods. Based on the robust principal component analysis model, Zhang et al. [20] proposed a classical low-rank matrix recovery (LRMR) model for HSI mixture noise removal. Given that the regularization terms can be easily integrated with each other, better denoising results can be obtained by combining low-rank and total variation, such as the total variation regularized low-rank matrix factorization (LRTV) [39] model. However, these low-rank models only convert the raw tensor of HSI into a matrix and do not take into account the precise spatial–spectral structure of HSI. Following this line, the low-rank tensor approximation methods have flourished in recent years; for example, Wang et al. [40] combined a tensor-based low-rank model with total variation and proposed the total variation regularized low-rank tensor decomposition (LRTDTV) model; Chang et al. [41] proposed the hyper-Laplacian regularized unidirectional low-rank tensor recovery (LLRT) model; and He et al. [42] proposed the nonlocal meets global (NGmeet) model.

B. Learning-Based HSI Denoising

In recent years, the explosion of deep learning has provided a new mindset for various applications in remote-sensing fields, such as super-resolution, denoising, and classification [3], [8], [14], [31]. Different from modeling approaches that extract prior information through manually designed regularization models, deep learning approaches extract the inherent multiscale features of HSIs through an end-to-end trainable way, which is learning the nonlinear mappings between noisy and clean HSIs automatically.

Hence, many deep learning methods have been proposed for HSI denoising. Xie and Li [43] used CNNs for the first time to remove noise from HSIs. Chang et al. [44] built HSI-DeNet which can be regarded as directly learning the multichannel 2-D filters without damaging the spectral–spatial structures. Yuan et al. [8] proposed the spatial–spectral deep CNN (HSID-CNN) model through multiscale feature extraction and multilevel feature representation by taking the joint spatial band and its adjacent correlated spectral bands into the residual CNN. Meanwhile, Liu and Lee [45] proposed a 3-D atrous denoising CNN (3DADCNN) using the 3-D atrous convolution layer. Dong et al. [11] utilized the modified U-Net [46] and built a 3-D U-Net network for HSI denoising. Maffei et al. [47] proposed HSI single denoising CNN (HSI-SDeCNN) by taking the noise-level map as input to train a network that is adaptive to multilevel noise without the need to train different models for different noise levels. Moreover, Wei et al. [48] proposed an alternating directional 3-D quasi-recurrent neural network (QRNN3D) for HSI denoising. By training the network on simulated HSI datasets, these deep learning methods achieved promising results for HSI denoising.

C. Visual Transformer

Influenced by the success of the transformer in the field of NLP, the application of the transformer in computer vision tasks has been proposed intensively in the last year along with excellent results. Vaswani et al. [32] proposed the transformer for machine translation work. Since then, the transformer has become a state-of-the-art method for various NLP tasks. The transformer is a new type of neural network structure for learning long-range dependencies, which makes learning global contextual information easier. In the field of computer vision, the vision transformer (ViT) was first proposed by Dosovitskiy et al. [33] and was used to undertake image classification tasks. Similar to tokens (words) in the transformer of NLP, ViT divides an image into small patches and feeds the linear embedding sequences of the patches into the transformer as tokens. To alleviate the reliance on large datasets in ViT, a data-efficient image transformer (DeiT) was developed by Touvron et al. [49], which uses data augmentation, token-based distillation, and teacher–student strategies to make the model efficient.

Given that ViT has a fixed resolution throughout the network, which neglects the fine-grained features [50], difficulties for various dense prediction tasks persist, thus leading to the paradigm of a hierarchical transformer being proposed followed by hierarchical CNNs. A typical hierarchical transformer is the pyramid vision transformer (PVT) backbone proposed by Wang et al. [34], which uses a gradually shrinking pyramid to reduce the computational workload of large feature maps to accommodate dense prediction tasks. Another hierarchical transformer backbone is the Swin transformer [29] introduced by Liu et al., which employs patch partition and patch merging operations to construct hierarchical feature maps and adopts a shifted windowing scheme to model global and boundary features.

Presently, the transformer has been successfully extended in low-level vision and many HSI tasks. For HSI processing tasks, He et al. [51] exploited a transformer structure with dense connection to capture spectral correlations while extracting spatial features by convolution for HSI classification. Ghosh et al. [52] combined convolutional autoencoder and transformer structures for hyperspectral unmixing. Wang et al. [53] utilized a spectral transformer encoder structure to extract spectral sequence information for HSI change detection. As for low-level vision tasks, Yang et al. [54] proposed a texture transformer network for image super-resolution (TTSR) by formulating low-resolution images and high-resolution images as queries and keys in the transformer. Meanwhile, Chen et al. [55] proposed the image processing transformer (IPT) to fully utilize the advantages of transformers by using large-scale pretraining and achieve the best performance in many image processing tasks including super-resolution, denoising, and deraining. Wang et al. [56] proposed the Uformer, which enhances the information extraction of local windows and employs a U-shaped transformer structure for image denoising.

Although the transformer has been extended in several image processing fields, it is still in the preliminary stage of exploration for low-level vision tasks, and related work remains rare. Moreover, the high computational cost of the transformer has caused considerable resistance to related research especially for HSI processing, as the computational complexity of self-attention on an image is quadratic to its size, which is difficult for the transformer to handle on HSI cubes. How to build a simpler and more efficient transformer

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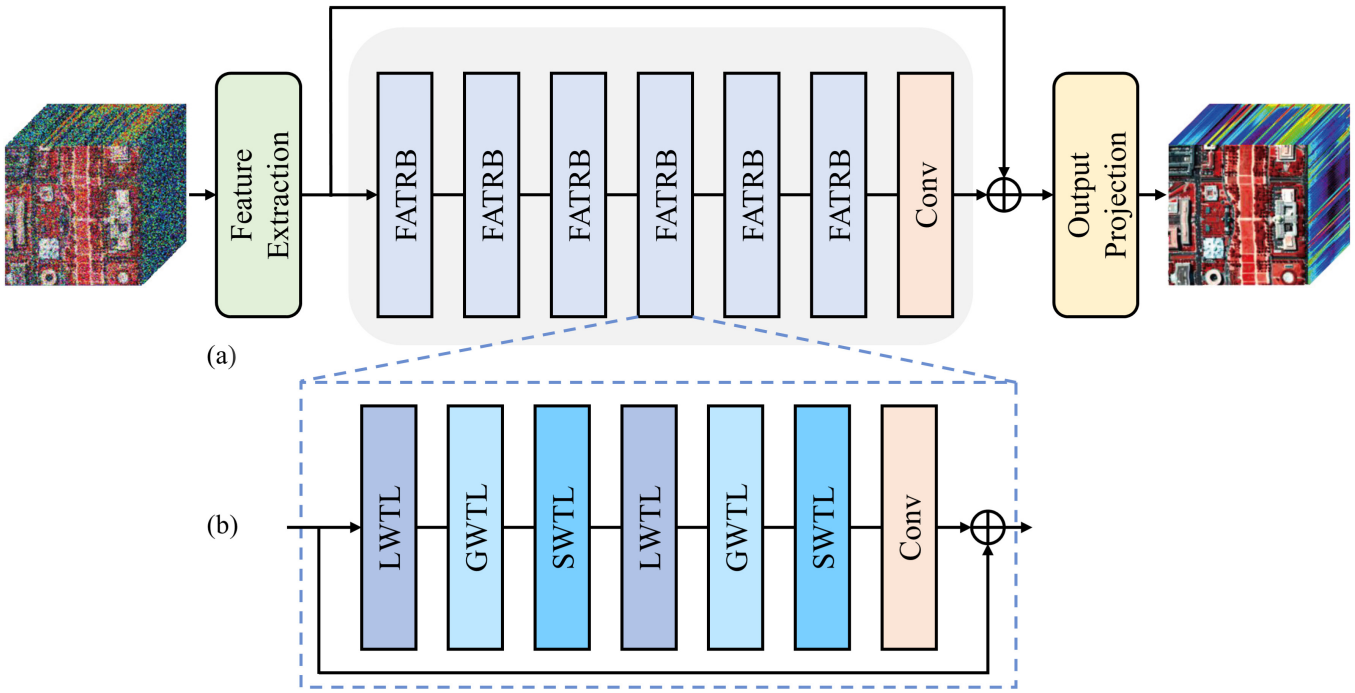

Fig. 1. Architecture of local–global feature-aware transformer based residual network (FATR): (a) overall architecture and (b) local–global feature-aware transformer based residual block (FATRB). In FATRB, LWTL denotes local windows transformer layer, GWTL denotes global windows transformer layer, and SWTL denotes shifted windows transformer layer.

structure or integrate it with existing convolutional structures remains to be further explored.

III. Proposed Model

A. Local–Global Feature-Aware Transformer Based Residual Network

The overview of the network structure proposed in this article is shown in Fig. 1, which contains three modules for processing HSIs. In this model, we first build a spatial–spectral feature extraction module to extract spatial and spectral shallow features. In this feature extraction module, several bands of HSI will be divided into small patches of spectral sequences and each patch is treated the same way as a token (word). Then, spatial and spectral raw features will be projected to a designed dimension using a spectral embedding layer.

After raw features projection in the previous feature extraction module, we use the deep feature extraction module to extract the deeper features of the HSI. Local–global feature-aware transformer based residual block (FATRB) can further excavate useful contextual information as well as local and global information by connecting three different types of window-based transformer layers. In the meanwhile, we use spatial-lightweight operation during self-attention in these blocks to reduce computational and memory costs, and we also use residual connections to ensure training stability. Additionally, we put a convolution layer at the end of the block to augment the features. Following that, by adopting six of these residual blocks (FATRBs), a deep feature extraction module is, thus, formed to extract hierarchical features. Finally, we use the reconstruction module to rebuild HSI from features. Different features from both branches of spatial–spectral feature extraction module and deep feature extraction module are merged into the final restoration to achieve a high-quality HSI reconstruction.

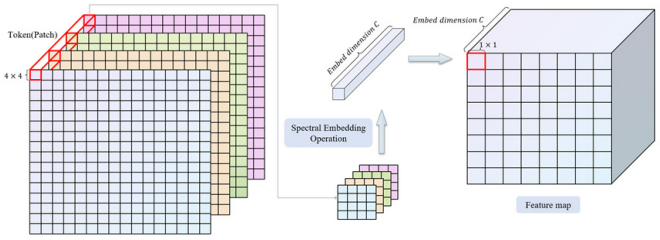

Fig. 2. Illustration of spectral embedding operation in spatial–spectral feature extraction module.

B. Spatial–Spectral Feature Extraction

As the HSI cubes are usually of high correlation and similarity in terms of surface properties and texture features, the redundant spectral information in HSI can greatly improve the accuracy of restoration. Therefore, for better utilizing and mining the character of high correction of adjacent bands, the proposed method introduces the patch embedding layer in the spatial–spectral feature extraction module. Different from CNN denoising, which uses convolution filters to extract feature maps, the transformer network uses linear embedding operations to transform the original data linearly. Here, we use a spectral embedding layer to extract the shallow features of the HSI. We first select a subcube $\boldsymbol{I}_k$ of size $H \times W \times 4$ from the original HSI data $\boldsymbol{I}_{\text{in}} \in \mathbb{R}^{H \times W \times B}$ as input data, where k denotes the k-th subcube of $\boldsymbol{I}_{\text{in}}$. We use a patch size of 4×4 as tokens. Then, we divide the input data $\boldsymbol{I}_k$ into $(H/4) \times (W/4)$ patches. Thus, the feature dimension of each patch is $4 \times 4 \times 4$. As shown in Fig. 2, each spectral patch is flattened and projected to a C-dimensional embedding of size $1 \times 1 \times C$. These embeddings can be used as feature

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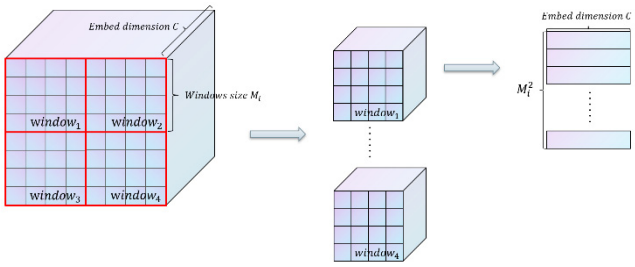

Fig. 3. Operation of windows partition before feature map is fed to MSA.

maps to extract deeper features further. After the spectral embedding, the size of the feature map is transformed to $(H/4) \times (W/4) \times C$. With such a joint spatial–spectral learning strategy, the module can simultaneously utilize the high correlation and complementarity of adjacent bands.

In this manner, the spatial and spectral information is initially extracted to form a feature map, after which subsequent subcubes are taken out in a specified stride starting from the current spatial–spectral cube until the entire HSI is traversed. Based on this subcube processing strategy, a clear advantage is that the amount of computational effort and memory cost can be significantly reduced, thus avoiding a shortage of memory due to the processing of the entire HSI. Moreover, the extraction of spatial–spectral features allows subsequent deep feature extraction modules to make use of the spectral information, thus avoiding spectral distortion of the restored HSI.

C. Spatial-Lightweight Window-Based Transformer Layer

In original windows multihead self-attention (W-MSA), given an input feature map with a size of $(H/4) \times (W/4) \times C$, we divide the input into different windows for separate processing. Each window is $M_i \times M_i$ in size, where i represents the i-th transformer layer. Then, the standard self-attention is computed separately for each window of size $M_i \times M_i \times C$, and the number of windows is $(HW/16M_i^2)$. Afterward, the window is reshaped to a $M_i^2 \times C$ feature, as shown in Fig. 3. For the feature $X \in \mathbb{R}^{M_i^2 \times C}$, as shown in Fig. 4, the query, key, and value matrices Q, K, and V are linearly transformed by

$$\begin{aligned} Q &= XW^Q \\ K &= XW^K \\ V &= XW^V \end{aligned} \tag{2}$$

where $Q, K, V \in \mathbb{R}^{M_i^2 \times d}$ and d denote the dimension of the query, key, and value matrices, respectively. $W^Q, W^K, W^V \in \mathbb{R}^{C \times d}$ are the linear projection matrices, which are shared among different windows. Then, the matrix of outputs is originally computed by the self-attention mechanism as

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^{\mathrm{T}}}{\sqrt{d}}\right)V. \tag{3}$$

In HSI denoising, to alleviate the computational and memory overhead, we use a spatial-lightweight strategy to

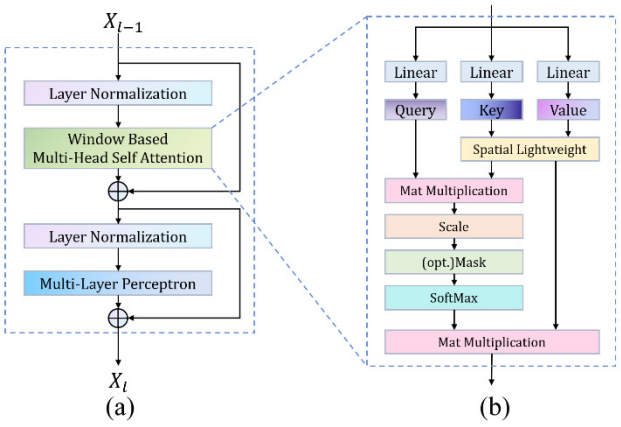

Fig. 4. (a) Transformer layer. (b) Specific calculation processing in spatial-lightweight W-MSA, (opt.) Mask denotes that a masking mechanism is employed in the shifted window-based MSA to restrict the computation of self-attention within each subwindow to avoid calculating feature maps that are not adjacent.

reduce the spatial size of K and V before the attention operation, i.e., $K' = SL(K) \in \mathbb{R}^{(M_i^2/k^2) \times d}$ and $V' = SL(V) \in \mathbb{R}^{(M_i^2/k^2) \times d}$. SL() is a function to reduce the spatial size of the input matrix K or V, denoted as

$$\text{SL}(Y) = \text{Norm}\big(\text{Reshape}(Y, k)W^{\text{S}}\big) \tag{4}$$

where $Y \in \mathbb{R}^{M_i^2 \times d}$ denotes the input matrix and k represents the scaling of reduction. Reshape(Y, k) is a function to reshape the input matrix Y into a sequence of size $(M_i^2/k^2) \times k^2 d$. $W^{\text{S}} \in \mathbb{R}^{k^2 d \times d}$ is the linear projection matrix for reducing the dimension of the input sequence back to d. Norm() refers to the layer normalization (LN) [57]. Based on the above operations, we then follow [29], [32], and [58] by adding a relative position bias B in computing our self-attention operation:

$$\text{Attention}\left(Q, K', V'\right) = \text{Softmax}\left(\frac{QK'^{\mathrm{T}}}{\sqrt{d}} + B\right)V' \tag{5}$$

where $B \in \mathbb{R}^{M_i^2 \times (M_i^2/k^2)}$ is the learnable encoding and randomly initialized. Then, we execute the attention function h times in parallel and concatenate the results of multi-head self-attention (MSA) as

$$\begin{aligned} \text{MSA}(Q, K, V) &= \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \\ \text{head}_i &= \text{Attention}\left(XW_i^Q, \text{SL}\left(XW_i^K\right), \text{SL}\left(XW_i^V\right)\right) \end{aligned} \tag{6}$$

where $W^O \in \mathbb{R}^{hd_{\text{head}} \times d}$ denotes the linear projection parameters, h is the head number of the self-attention operation, $W_i^Q, W_i^K, W_i^V \in \mathbb{R}^{C \times d_{\text{head}}}$ are the i-th projection matrices, and the dimension of the i-th head d_{head} is equal to (C/h).

In the next step, to transform the features extracted from MSA further, a multilayer perceptron (MLP) which has two fully connected layers before and after as well as a Gaussian error linear unit (GELU) nonlinearity activation [59] in between is used. In addition, an LN layer is used before each MSA and each MLP, and a residual connection is followed

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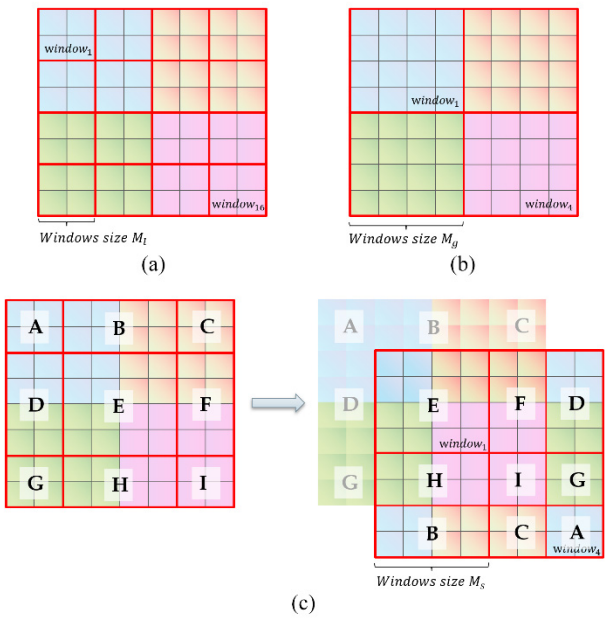

Fig. 5. Illustration of the window partitioning approach of feature maps for computing self-attention: (a) local windows pattern; (b) global windows pattern; and (c) shifted windows pattern.

by both MSA and MLP. The overall processing flow of a transformer layer can be formulated as

$$\begin{aligned} \hat{X}^l &= \mathrm{MSA}\big(\mathrm{LN}\big(X^{l-1}\big)\big) + X^{l-1} \\ X^l &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{X}^l\big)\big) + \hat{X}^l \end{aligned} \tag{7}$$

where $\hat{X}^l$ and X^l denote the output features from MSA and MLP modules, respectively, and LN denotes the layer normalization.

D. Local–Global Feature-Aware Transformer Based Residual Block

The original window-based self-attention module divides the feature map into static windows, and only local information of features can be perceived in these small windows, which limits its modeling capabilities. In order to extract more useful information, we introduce cross-window connections through the shifted windows partitioning pattern from Swin transformer [29]. In this article, we built three types of different transformer layers in a FATRB, i.e., local windows, global windows, and shifted windows transformer layer (SWTL).

Among these layers, the difference lies in the way in which the windows are partitioned. The local windows transformer layer (LWTL) divides the feature map into more small-sized windows to feed into the layers, while the global windows transformer layer (GWTL) divides the feature map into fewer large-sized windows to further enhance the extraction of features. In addition, the shifted windows can extract information across windows, which work in conjunction with the GWTL to obtain a larger range of global features. The division of the window is shown in Fig. 5, where M_{l}, M_{g}, and M_{s} represent the sizes of local windows, global windows, and shifted windows, respectively.

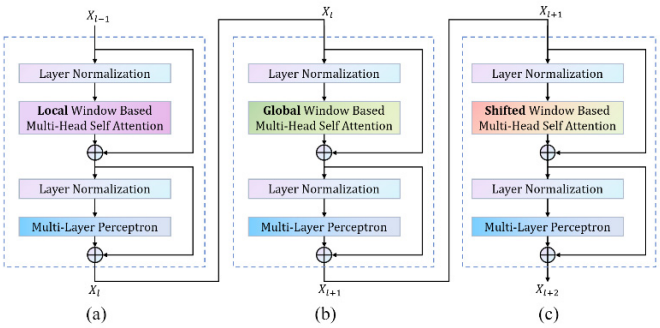

Fig. 6. Three consecutive transformer layers: (a) LWTL; (b) GWTL; and (c) SWTL.

Then, we stack several window-based transformer layers to extract and transform the features at a deep level, three consecutive layers are illustrated in Fig. 6, and consecutive layers are computed as

$$\begin{aligned} \hat{X}^l &= \text{LW-MSA}\big(\mathrm{LN}\big(X^{l-1}\big)\big) + X^{l-1} \\ X^l &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{X}^l\big)\big) + \hat{X}^l \\ \hat{X}^{l+1} &= \text{GW-MSA}\big(\mathrm{LN}\big(X^l\big)\big) + X^l \\ X^{l+1} &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{X}^{l+1}\big)\big) + \hat{X}^{l+1} \\ \hat{X}^{l+2} &= \text{SW-MSA}\big(\mathrm{LN}\big(X^{l+1}\big)\big) + X^{l+1} \\ X^{l+2} &= \mathrm{MLP}\big(\mathrm{LN}\big(\hat{X}^{l+2}\big)\big) + \hat{X}^{l+2} \end{aligned} \tag{8}$$

where $\hat{X}^l$ and X^l denote the output features from W-MSA and MLP module for layer l, respectively, LN denotes the layer normalization. LW-MSA, GW-MSA, and SW-MSA denote the W-MSA using local windows, global windows, and shifted windows partitioning setups, respectively.

After two successive sets of three different layers, a convolutional layer is employed for aggregation. Afterward, we add a residual connection at the end to ensure the stability and accuracy of the training process. Finally, we form a FATRB, as shown in Fig. 1(b).

IV. Experiments

A. Implementation Details

In order to validate the performance of the method proposed in this article for restoring HSI from noises, synthetic and real-world HSI data will be used in the experiments, respectively. In the meantime, we compare the proposed method with 12 different model-based and deep learning-based methods, including a filtering-based approach BM4D [27], a total variation method E3DTV [38], a matrix-based low-rank method LRMR [20], two total variation regularized total variation matrix-based methods LRTV [39] and local low-rank matrix recovery and global spatial–spectral total variation (LLRGTV) [60], two complex noise modeling methods non-i.i.d. mixture of Gaussians (NMoGs) noise assumption [61] and fast hyperspectral denoising (FastHyDe) [62], two tensor-based total variation methods LRTDTV [40] and NGmeet [42], three deep learning methods HSID-CNN [8], HSI-SDeCNN [47], and QRNN3D [48].

1) Datasets: The following six datasets were used in the experiments, with the grayscale values of the HSIs in each band being independently normalized.

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1) The first dataset is the Washington DC Mall (WDC) image with the size of 1280 × 303 × 191, which was acquired by the Hyperspectral Digital Imagery Collection Experiment (HYDICE) airborne sensor. The entire data were cropped to 1024 × 303 × 191 for training and the rest to 256 × 256 × 191 for use to test in the synthetic experiments, respectively.
2) The second dataset is the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) Indian Pines Test Site3 dataset with a size of 145 × 145 × 220, which was used for real data experiments.
3) The third dataset is the Urban dataset acquired by the HYDICE sensor, which retains all the original bands with a size of 307 × 307 × 210 for real data experiments.
4) The fourth dataset is the Earth Observing-1 (EO-1) Hyperion dataset with a size of 400 × 200 × 166, which was used in the real data experiments after removing the water absorption bands.
5) The fifth dataset is the Zhuhai-1 dataset in the area of Chaohu, Anhui, China. This image was cropped to 400 × 400 × 32 for testing in the real data experiments.
6) The sixth dataset is the Gaofen-5 (GF-5) dataset in Dunhuang, Gansu, China. The image was cropped to 400 × 400 × 150 for testing in the real data experiments.

2) Quantitative Metrics: For an accurate evaluation of the experimental performance, the peak signal-to-noise ratio (PSNR), the structural similarity (SSIM) [63], the spectral angle mapper (SAM) [64], relative dimensionless global error in synthesis (ERGAS) [65], and the feature similarity index (FSIM) [66] are used as quality assessment metrics in the synthetic experiments. The PSNR and SSIM evaluate the similarity between the denoised image and the original clean image by mean square error and structural consistency, respectively. FSIM assesses the similarity of images by the phase congruency and image gradient magnitude. SAM calculates the angular distance between the spectrum vectors of the target and reference images over all spatial locations, and it forms a map matrix of the full image spatial size calculated by all spectral pairs. ERGAS measures the fidelity of the denoised image calculated from the weighted sum of mean square error in each band. For the HSIs, we calculate the mean PSNR (MPSNR), mean SSIM (MSSIM), and mean FSIM (MFSIM) among all bands, and the mean spectral angle distance (MSAD) across the whole calculated map matrix. Remarkably, the higher MPSNR, MSSIM, and MFSIM values reveal the better spatial quality after restoration, and the lower MSAD and ERGAS mean the better spectral quality of the recovered HSI.

3) Noise Setting of Synthetic Experiments: In synthetic experiments, four cases of corruptions are added to the WDC test data as follows.

1) *Case 1 (i.i.d. Gaussian Noise)*: Most of the existing model-based and learning-based HSI denoising methods are validated by simulating independent identically distributed (i.i.d.) additive white Gaussian noise in different bands, which means that the noise intensity is the same in different bands. For a fair comparison, we simulated zero-mean Gaussian noise in each band according to the same distribution, with the standard deviation of Gaussian noise set to $\sigma = \{25, 50, 75, 100\}$, respectively, in four subcases (Cases 1.1–1.4).
2) *Case 2 (Non-i.i.d. Gaussian Noise)*: Real-world HSIs are often disturbed by unknown random noise; hence, we simulated non-i.i.d. noise, where the noise intensity varies among bands. In this case, the Gaussian noise σ for each band is randomly selected from 0 to 25.
3) *Case 3 (Gaussian Noise + Stripe Noise)*: All bands were corrupted by non-i.i.d. Gaussian noise as Case 2. Moreover, 60 bands in WDC data were randomly selected to add stripe noise and the number of stripes in each band is from 20 to 50.
4) *Case 4 (Gaussian Noise + Impulse Noise)*: Each band was disturbed by Gaussian noise as in Case 2. In addition, impulse noise with a noise intensity of 1% was added to each band.
5) *Case 5.1 (Gaussian Noise + Stripe Noise + Impulse Noise)*: The non-i.i.d. Gaussian noise in Case 2, random stripe noise in Case 3, and impulse noise in Case 4 were mixed to contaminate the HSIs.
6) *Case 5.2 (Gaussian Noise + Stripe Noise + Impulse Noise)*: The i.i.d. Gaussian noise in Case 1.2, random stripe noise in Case 3, and impulse noise in Case 4 were mixed to contaminate the HSIs.

B. Training Details

1) Loss Function: To optimize our network, we use the Charbonnier penalty function [67], [68], [69] as our loss function, which can be denoted as

$$\mathcal{L} = \sqrt{\|I_{\mathrm{RHSI}} - I_{\mathrm{LHSI}}\|^2 + \epsilon^2} \tag{9}$$

where I_{RHSI} is the predicted HSI after restoration from the network, I_{LHSI} is the ground-truth label HSI, and ϵ is a constant parameter that is empirically set to 10^{-3}. With the guidance of the loss function, our network can be trained in a supervised manner so that the predicted clean HSI is maintained as similarly as possible to the ground-truth HSI.

2) Implementation Strategy: We use the Adam [70] algorithm as the gradient descent optimization method for training, where $\beta_1 = 0.9$ and $\beta_2 = 0.999$, and the learning rate is initialized to 2×10^{-4}. The batch size is set to 4, which means that we train four HSI samples at one time, and we randomly crop the training data with a size of 1280 × 303 × 191 into 128 ×128 × 191 patches as input. In addition, we use 1% and 2% linearization, rotating, and flipping to augment our data. All the trained models are training from WDC data. For the synthetic experiments, the individual cases are trained based on the noise settings of the corresponding cases. For the real experiments, the noise setting of the model under training is similar to Case 5.1. The difference from Case 5.1 is that Gaussian noise with $\sigma = 50$ is added for a quarter of the training data, $\sigma = 100$ is added for a quarter of the training data, and non-i.i.d. noise is added for half of the training data same as in Case 5.1. In addition, half of the bands are randomly selected to add stripe noise in all the training

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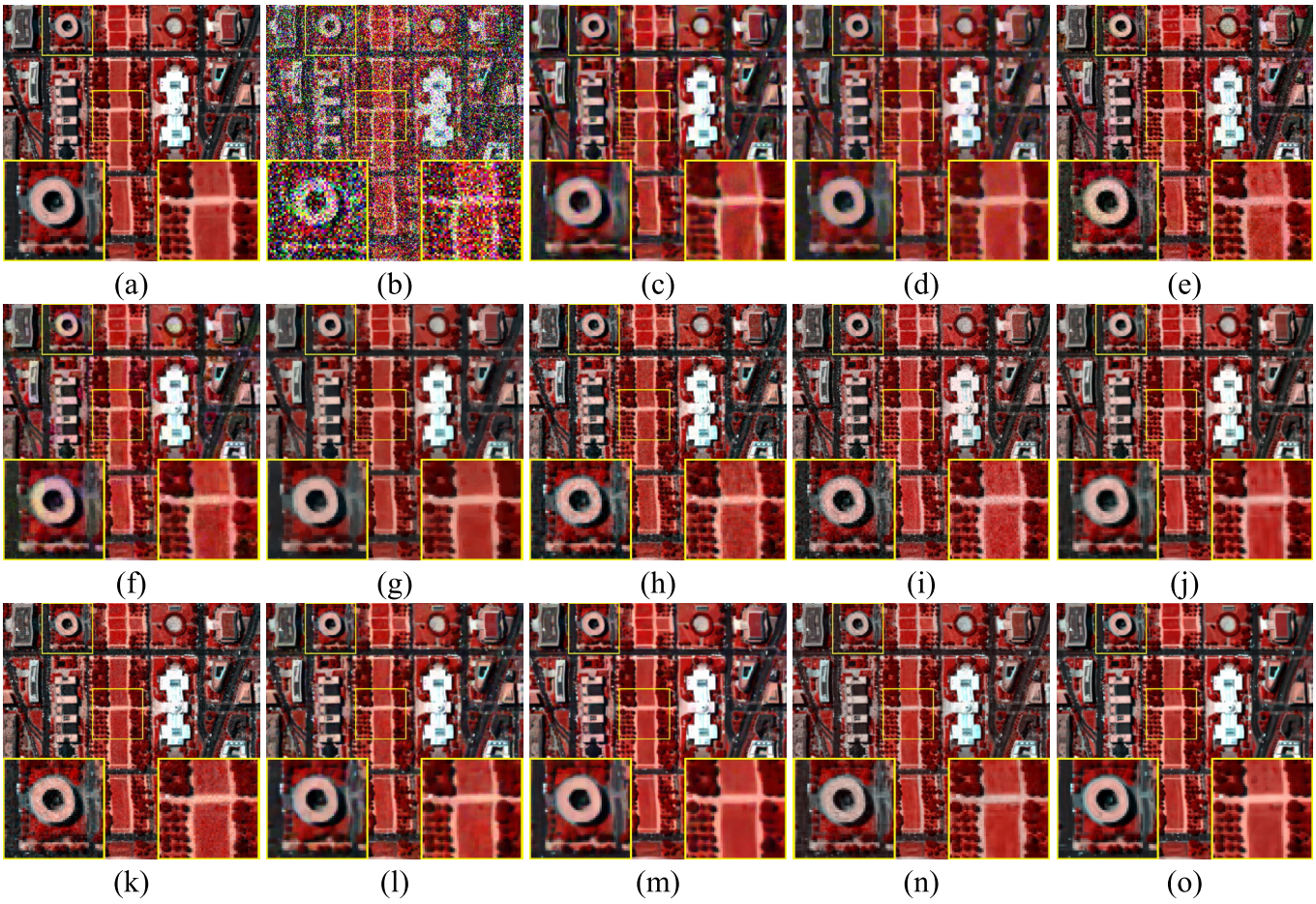

Fig. 7. Synthetic Gaussian noise with noise level $\sigma = 100$ denoising comparison in Case 1.4, false color image with bands (57, 27, 17): (a) original image; (b) noisy image; (c) BM4D; (d) LRTV; (e) LRMR; (f) LLRGTV; (g) E3DTV; (h) LRTDTV; (i) NMoG; (j) FastHyDe; (k) NGmeet; (l) HSID-CNN; (m) HSI-SDeCNN; (n) QRNN3D; and (o) FATR.

data, and the rest of the settings are consistent with Case 5.1. In the network, we set the size of the local windows to 8×8, the size of the global window is set to 16×16, the head number is set to 6, and the embedding dimension C is set to 192. The training model framework is implemented in PyTorch 1.7.1 based on CUDA 10.1 library and accelerated on four NVIDIA RTX 2080Ti GPUs. A total of 5000 epochs had to be undergone to train our model until the loss converged, with 100 iterations per epoch and about 1.2–1.8 min per epoch, taking about five–seven days for the entire training.

C. Synthetic Experiments

The visual effects of different comparison methods on the WDC test dataset in Case 1.4 ($\sigma = 100$) and Case 5.2 (mixture noise case) are selected for demonstration, corresponding to Figs. 7 and 8, respectively. Since HSI has a large volume of bands, only a few selected bands (R:57, G:27, B:17) are displayed in false color in Figs. 7 and 8. To visualize the spatial detail of the recovered HSI better, some local areas of the false color displayed image have been enlarged and are presented in the bottom part of Figs. 7 and 8. The original and noisy images are shown in Fig. 7(a) and (b) of Case 1.4 and in Fig. 8(a) and (b) of Case 5, respectively. The denoising results of BM4D, LRTV, LRMR, LLRGTV, E3DTV, LRTDTV, NMoG, FastHyDe, NGmeet, HSID-CNN, HSI-SDeCNN, QRNN3D, and our FATR are shown in Fig. 7(c)–(o) of Case 1.4 and in Fig. 8(c)–(o) of Case 5.2, respectively.

For Case 1.4, in the condition of extremely heavy Gaussian noise, BM4D removes most of the Gaussian noise but causes significant over-spatial distortion. Although LRTV and E3DTV are capable of eliminating noise, they also lead to over-smoothing results. LRMR, LRTDTV, NMoG, and NGmeet preserve more spatial details and edge information while leaving some residual noise without being completely removed. LLRGTV removes most of the noise, but spatially discontinuous distortion occurs in some areas. FastHyDe, HSID-CNN, HSI-SDeCNN, and QRNN3D are generally better at denoising, with FastHyDe, HSID-CNN, and QRNN3D retaining the most textural detail but with a small amount of noise not removed and with HSI-SDeCNN yielding over-smoothing. Our proposed FATR removes severe noise while retaining local information and textural detail, which is superior to other methods by a wide margin.

For Case 5.2 with a mixture of Gaussian noise, stripes, and impulse noise, BM4D cannot remove the stripes and impulse noise. LRTV, LLRGTV, and E3DTV methods

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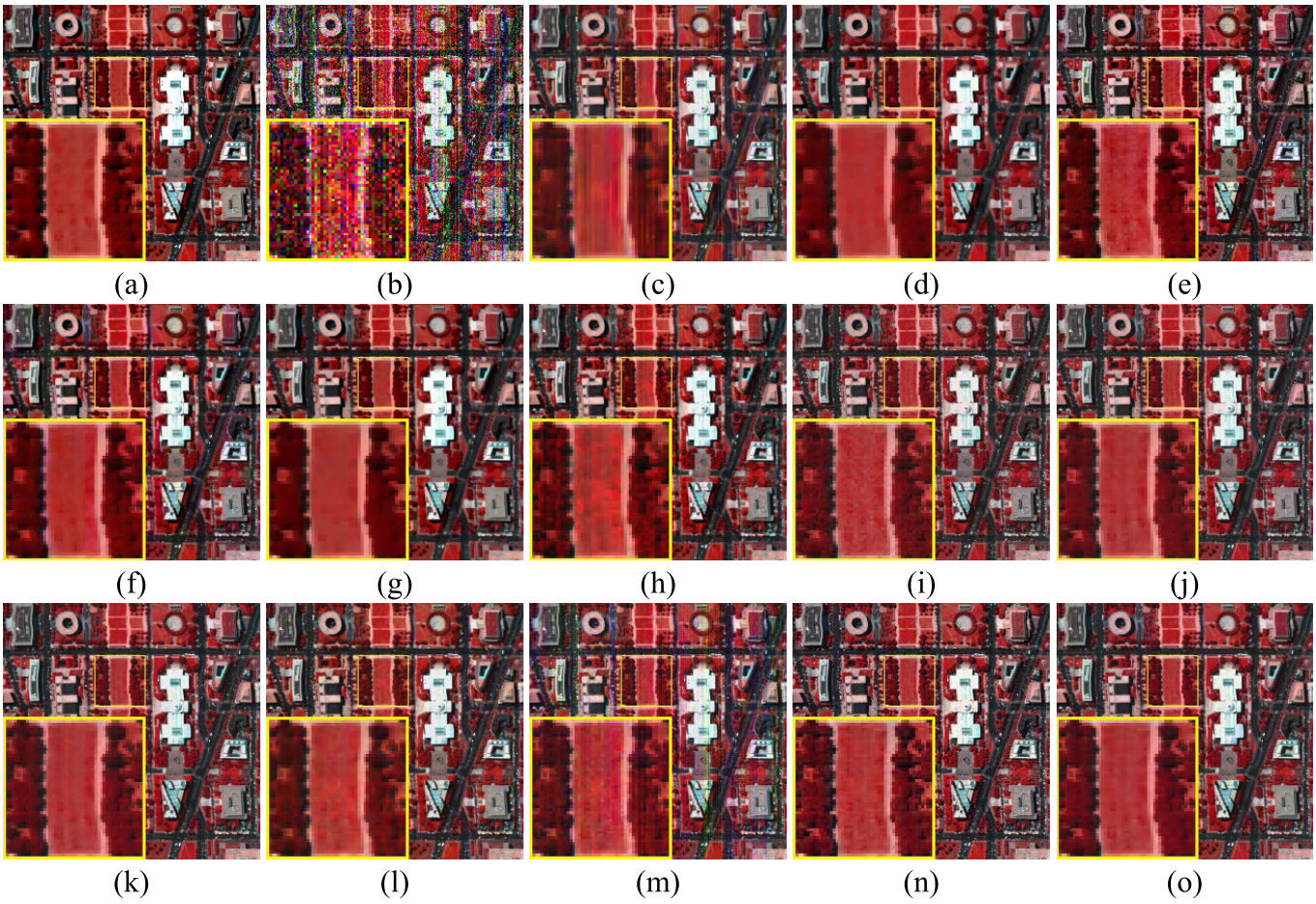

Fig. 8. Synthetic hybrid noise reduction results comparison in Case 5.2, false color image with bands (57, 27, 17): (a) original image; (b) noisy image; (c) BM4D; (d) LRTV; (e) LRMR; (f) LLRGTV; (g) E3DTV; (h) LRTDTV; (i) NMoG; (j) FastHyDe; (k) NGmeet; (l) HSID-CNN; (m) HSI-SDeCNN; (n) QRNN3D; and (o) FATR.

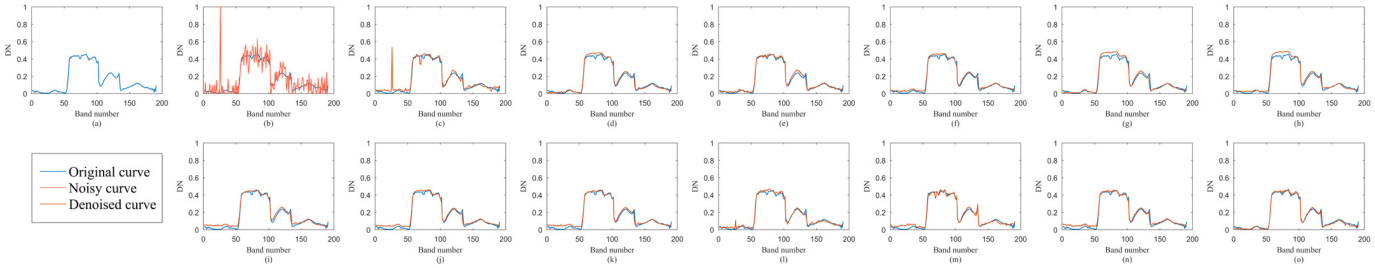

Fig. 9. DN value curves in different bands of pixel (70, 140) after restored from synthetic hybrid noise of Case 5.2: (a) original image; (b) noisy image; (c) BM4D; (d) LRTV; (e) LRMR; (f) LLRGTV; (g) E3DTV; (h) LRTDTV; (i) NMoG; (j) FastHyDe; (k) NGmeet; (l) HSID-CNN; (m) HSI-SDeCNN; (n) QRNN3D; and (o) FATR.

produce over-smoothing results with losing spatial detail and edge features after denoising. LRMR, LRTDTV, NMoG, and FastHyDe methods maintain more textural details and spatial features after denoising while leaving some residual Gaussian noise. NGmeet, HSID-CNN, HSI-SDeCNN, and QRNN3D have a small amount of shallow stripe and Gaussian residual noise at the grass location in the middle of the image, where HSI-SDeCNN has a number of evident stripes in the right-hand building area that cannot be removed. Still, the proposed FATR can eliminate hybrid noise greatly and maintain an excellent level of spatial textural detail simultaneously.

In addition to the visual evaluation of the spatial domain, we have also selected typical ground objects to show their recovered spectra after denoising to compare the capabilities of different methods in terms of spectral restoration. We selected the pixel of the tree class at position (70, 140) on the WDC test data of Case 5.2, and the recovered spectral curves of different methods are presented in Fig. 9, where the x- and y-axes are the band number and the normalized digital number (DN) value, respectively. The noisy image spectrum shown in orange curve in Fig. 9(b) exhibits significant jagged distortion compared with the original blue curve in Fig. 9(a), with the

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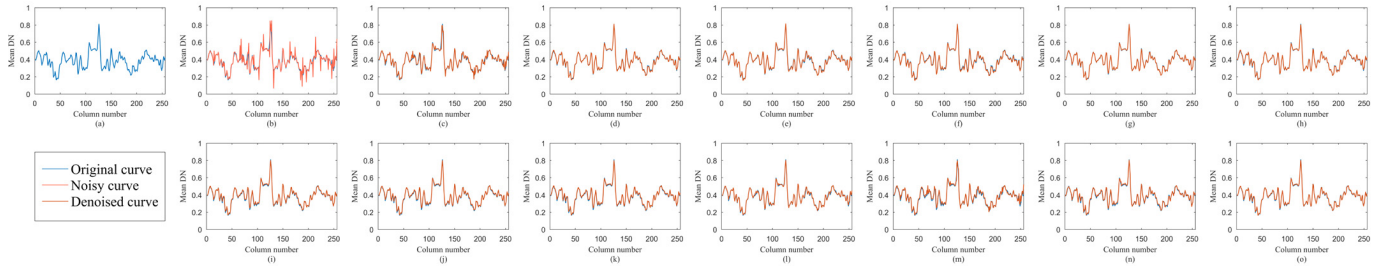

Fig. 10. Synthetic hybrid noise reduction results of Case 5.2 displayed in vertical mean DN profiles of band 165: (a) original image; (b) noisy image; (c) BM4D; (d) LRTV; (e) LRMR; (f) LLRGTV; (g) E3DTV; (h) LRTDTV; (i) NMoG; (j) FastHyDe; (k) NGmeet; (l) HSID-CNN; (m) HSI-SDeCNN; (n) QRNN3D; and (o) FATR.

spectral curve fluctuating up and down. Fig. 9(c)–(o) shows the recovered spectra of each method in red line. As demonstrated, our proposed FATR can recover the spectral curve better than other comparison methods without causing distortion.

We also display the longitudinal profile of Case 5.2, which can be used to demonstrate the spatial recovery for stripe noise. In particular, Fig. 10 plots the longitudinal profile of the WDC test data in the band 165 of Case 5.2, where the x- and y-axes are the column number and the mean DN value for each column, respectively. The orange curve in Fig. 10(b) shows that the mean DN of the columns with stripe noise deviates significantly from the original blue curve in Fig. 10(a). The red line contours restored by the different methods are compared with the blue line of the clean image in Fig. 10(c)–(o). The restored profile by our proposed FATR method most closely resembles to the original HSI, which suggests that our method can eliminate the spatial distribution of stripe noise robustly. Based on the aforesaid results, we deduce that our proposed method can remove hybrid noise as well as severely Gaussian noise and maintain the relative restoration of the spectra without distortion.

The restoration results achieved by different algorithms on the WDC data in Case 1.1–Case 5.2 are presented in Table I, respectively. In addition, the computational costs are also presented in Table I. It can be seen that our proposed method achieves the best quantitative evaluation indicator results in these five different cases. In particular, the spectral recovery results of our methods, such as MSAD and ERGAS, can achieve far better results than other comparison algorithms in most cases. Fig. 11 illustrates the bandwise PSNR, SSIM, and FSIM values for the quantitative evaluation in Case 1.4 and Case 2–Case 5.1 for the WDC test dataset. In most cases, our method is sufficient to maintain the highest indicator values in most bands, and only in a very rare case does it fail to outperform other methods. Thus, our method also provides a good result in dealing with spatial restoration in most bands.

D. Real Experiments

To evaluate the effectiveness of our proposed method further, five real-world HSI datasets were used for the real experiments, namely, Indian Pine Site3 dataset, Urban dataset, EO-1 dataset, Zhuhai-1 dataset, and GF-5 dataset.

1) AVIRIS Indian Pines Site 3 Dataset: The first few bands and several other bands of the Indian Pines HSI are seriously degraded by Gaussian noise and impulse noise [71]. The denoising results of the compared method and the proposed method are shown in Fig. 12, with the first band displayed as the grayscale image.

As can be seen in the comparison figure of the results, the original first band has a large amount of Gaussian noise and stripe noise. The methods BM4D, LRMR, HSI-SDeCNN, and QRNN3D can remove Gaussian noise but they seem to be almost powerless against heavy stripes. Meanwhile, LRTV, E3DTV, LRTDTV, NGmeet, and HSID-CNN can remove most of the Gaussian noise and coarse stripes, but there is still slight or insignificant stripe noise which is not completely removed. NMoG, FastHyDe, and our method are more effective in the visual result, removing both Gaussian noise and stripe noise while maintaining clear detail and structural information.

With Indian Pines Site3 dataset providing 16 classes of ground-truth labels for classification, we employed the support vector machine (SVM) [72] to classify the denoised HSIs for further validation of each method. Eventually, the overall accuracy (OA), average accuracy (AA), and kappa coefficients of the resulting classification maps were used for evaluation. The classification quantification results are summarized in Table II, which includes the results before and after denoising by different methods, respectively.

From a quantitative results perspective, our method demonstrates outperformance on the Indian Pine dataset. It is also evident in Fig. 13 of the classification visual results, where Fig. 13(a) is the ground-truth, Fig. 13(b) is the classification map obtained from the original noisy images, Fig. 13(c)–(n) are the results of the comparison methods, Fig. 13(o) is the result of our method, and Fig. 13(p) is the category description. These classification visual results indicate that the results of NGmeet, HSID-CNN, HSI-SDeCNN, QRNN3D, and our proposed FATR are less fragmented and more complete, while HSI-SDeCNN and our proposed method can also obtain better classification accuracy as listed in Table II.

2) HYDICE Urban Dataset: The raw Urban data are polluted by a variety of degradation problems in some of the bands, such as Gaussian noises, stripe noises, deadlines, atmosphere, water absorption, and other unknown noise [39]. The denoising results of different comparison methods and our method for the Urban data are presented in Fig. 14, where we

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TABLE I

QUANTITATIVE EVALUATION OF NOISE REDUCTION RESULTS IN THE SYNTHETIC EXPERIMENTS

Noise Case	Index	Noisy	BM4D	LRTV	LRMR	LLRGTV	E3DTV	LRTDTV	NMoG	FastHyDe	NGmeet	HSID-CNN	HSI-SDeCNN	QRNN3D	FATR
Case 1.1∼1.4: i.i.d. Gaussian Noise															
Case 1.1	MPSNR	20.1729	31.1460	30.1916	31.9333	31.2150	31.0716	31.8756	32.7188	32.8189	33.1895	32.2430	33.2804	33.3891	**33.4172**
	MSSIM	0.7392	0.9657	0.9599	0.9740	0.9720	0.9672	0.9709	0.9778	0.9796	0.9802	0.9730	0.9795	0.9821	**0.9823**
	MSAD	19.8456	6.3811	5.1066	6.0915	5.8808	5.2531	5.4029	5.6995	5.2791	5.3511	5.5460	4.8998	5.1460	**3.5138**
	ERGAS	259.79	74.39	82.59	71.68	76.58	80.45	69.07	66.98	65.78	63.03	67.29	59.43	59.98	**58.13**
	MFSIM	0.8761	0.9620	0.9542	0.9507	0.9514	0.9314	0.9581	0.961	0.963	0.9634	0.9650	**0.9766**	0.9664	0.9724
Case 1.2	MPSNR	14.1487	26.8189	26.5906	28.2670	28.7222	28.7947	29.1135	29.3422	29.438	29.5807	29.1090	29.7038	29.8465	**30.0249**
	MSSIM	0.4864	0.9161	0.9127	0.9413	0.9506	0.9482	0.9448	0.9517	0.9588	0.9503	0.9502	0.9578	0.9659	**0.9668**
	MSAD	32.4379	8.8141	7.1196	8.1653	7.0605	6.2827	7.6795	7.6303	7.0714	7.5699	7.2674	6.4357	6.3424	**4.6598**
	ERGAS	519.83	120.87	126.31	105.38	98.79	99.96	95.43	94.24	94.58	91.69	94.44	87.59	**82.13**	83.80
	MFSIM	0.7617	0.9235	0.9130	0.9206	0.9396	0.9277	0.9191	0.9327	0.9299	0.9223	0.9498	**0.9612**	0.9430	0.9552
Case 1.3	MPSNR	10.6311	24.3403	24.5671	25.7947	26.6957	26.1616	26.6426	26.9115	27.1800	27.0719	26.9351	27.4816	28.0269	**28.1912**
	MSSIM	0.3269	0.8610	0.8679	0.9041	0.9235	0.9094	0.9079	0.9223	0.9389	0.9196	0.9234	0.9334	0.9469	**0.9517**
	MSAD	41.4718	10.6417	8.4256	10.0748	8.3989	7.3251	10.0047	9.2930	8.2422	9.4400	8.5643	7.6338	7.7532	**5.4563**
	ERGAS	779.27	160.22	159.77	138.69	123.60	132.01	127.37	123.21	119.71	120.56	119.99	112.10	108.64	**106.32**
	MFSIM	0.6774	0.8894	0.8188	0.8926	0.9180	0.9049	0.8805	0.9070	0.9148	0.9047	0.9335	**0.9447**	0.9229	0.9347
Case 1.4	MPSNR	8.1304	22.6506	23.0909	23.9696	24.9289	24.3262	24.6566	25.0638	25.2052	25.2190	25.4953	25.9162	26.0574	**26.6115**
	MSSIM	0.2272	0.8056	0.8237	0.8657	0.8895	0.8698	0.8689	0.8917	0.9048	0.8900	0.8975	0.9083	0.9212	**0.9360**
	MSAD	48.2370	12.1532	9.5876	11.7554	9.8284	9.1140	12.0814	10.7991	9.2338	10.7219	9.4985	8.6502	9.7906	**5.9332**
	ERGAS	1039.39	194.36	189.12	169.89	151.01	161.90	158.31	151.50	147.24	143.40	140.94	133.70	137.84	**123.85**
	MFSIM	0.6130	0.8522	0.8514	0.8700	0.8971	0.8831	0.8459	0.8857	0.8922	0.8617	0.9185	0.9267	0.8905	**0.9278**
Case 2: Non-i.i.d. Gaussian Noise															
Case 2	MPSNR	29.0757	34.7117	33.5173	34.3798	32.6854	34.4177	32.8023	35.5757	35.7299	36.2468	37.1598	36.1783	37.0981	**38.7990**
	MSSIM	0.8913	0.9826	0.9818	0.9849	0.9804	0.9844	0.9769	0.9890	0.9898	0.9899	0.9891	0.9883	0.9913	**0.9940**
	MSAD	12.7258	4.9794	3.8663	4.9317	5.2563	4.1017	4.6950	4.4072	4.0514	4.0158	3.8179	3.9385	3.5323	**3.0208**
	ERGAS	147.73	51.97	57.78	56.54	67.30	58.34	62.02	55.49	46.92	44.58	41.48	44.48	40.61	**36.14**
	MFSIM	0.9046	0.9760	0.9704	0.9632	0.9570	0.9598	0.9689	0.9738	0.9741	0.9766	0.9812	0.9813	0.9769	**0.9815**
Case 3: Non-i.i.d. Gaussian Noise + Stripe Noise															
Case 3	MPSNR	27.2355	32.3312	33.0617	33.9081	32.478	34.1719	32.3101	35.4971	35.4253	35.2204	34.4797	34.4061	35.9233	**36.1674**
	MSSIM	0.8693	0.9669	0.9797	0.9835	0.9796	0.9837	0.9750	0.9892	0.9893	0.9887	0.9787	0.9833	0.9898	**0.9913**
	MSAD	14.3532	7.0841	4.1735	5.0856	5.3320	4.1221	4.6178	4.4207	4.3552	4.2589	5.7836	4.7887	4.0158	**3.9864**
	ERGAS	171.83	76.58	61.54	59.13	68.67	60.00	65.57	56.15	55.97	49.80	64.53	55.01	49.86	**49.75**
	MFSIM	0.9937	0.9796	0.9713	0.9649	0.9574	0.9592	0.9719	0.9769	0.9787	0.9753	**0.9887**	0.9812	0.9757	0.9783
Case 4: Non-i.i.d. Gaussian Noise + Impulse Noise															
Case 4	MPSNR	22.1772	29.1729	33.0495	34.0298	32.5239	34.2073	32.1455	33.9013	33.9546	34.5719	33.3685	33.4418	33.9481	**35.9603**
	MSSIM	0.8327	0.9511	0.9795	0.9831	0.9798	0.9839	0.973	0.9809	0.9833	0.9836	0.9755	0.9776	0.9826	**0.9901**
	MSAD	16.1648	7.828	4.1473	5.1945	5.3172	**4.1328**	4.8013	4.9551	4.8606	4.5152	5.4158	5.0627	4.7650	4.1688
	ERGAS	213.91	95.05	61.02	58.23	68.22	58.26	66.61	57.18	57.44	52.35	59.34	58.85	56.16	**49.48**
	MFSIM	0.9464	0.9621	0.9669	0.963	0.9564	0.9617	0.9698	0.9664	0.9688	**0.9731**	0.9712	0.9684	0.9685	0.9706
Case 5.1: Non-i.i.d. Gaussian Noise + Stripe Noise + Impulse Noise															
Case 5.1	MPSNR	20.192	28.9691	30.9835	32.3650	31.4097	32.0299	31.1704	31.6966	32.192	32.4587	31.3501	31.3102	32.2923	**33.0464**
	MSSIM	0.7578	0.9424	0.9668	0.9759	0.9739	0.9736	0.9658	0.9712	0.9753	0.9745	0.9661	0.9658	0.9748	**0.9796**
	MSAD	19.1681	7.9739	5.0116	5.9252	5.7524	4.9697	5.3449	6.0351	5.6426	5.5522	6.0982	6.2431	5.5211	**4.9475**
	ERGAS	261.83	97.26	75.61	68.66	75.32	72.63	74.43	73.74	69.89	66.14	73.57	75.20	67.59	**60.14**
	MFSIM	0.9109	0.9585	0.9611	0.9548	0.9531	0.9432	0.9647	0.9611	0.9642	0.9638	0.9658	0.9639	0.9653	**0.9682**
Case 5.2: i.i.d. Gaussian Noise + Stripe Noise + Impulse Noise															
Case 5.2	MPSNR	13.9931	24.6648	26.167	27.7488	27.1598	26.8561	27.6685	26.3043	26.9600	27.1323	26.3176	24.4509	26.6384	**28.0644**
	MSSIM	0.4603	0.8741	0.9043	0.9390	0.9389	0.9220	0.9277	0.9257	0.9381	0.9395	0.9168	0.8795	0.9337	**0.9439**
	MSAD	30.7564	9.6131	8.1471	7.2965	7.4083	7.4339	7.2967	8.4902	7.6499	7.5746	8.8113	10.4831	7.9030	**6.9160**
	ERGAS	528.36	155.79	132.29	111.34	118.91	122.23	111.00	131.52	121.72	119.12	129.74	161.81	127.64	**106.27**
	MFSIM	0.7765	0.9067	0.9187	0.9249	0.9376	0.9096	0.9370	0.9347	0.9424	0.9425	0.9368	0.9026	0.9277	**0.9455**
Time	Second	—	1416.05	332.30	770.34	328.67	328.67	515.83	191.28	**0.75**	186.87	4.92	2.97	3.69	15.85

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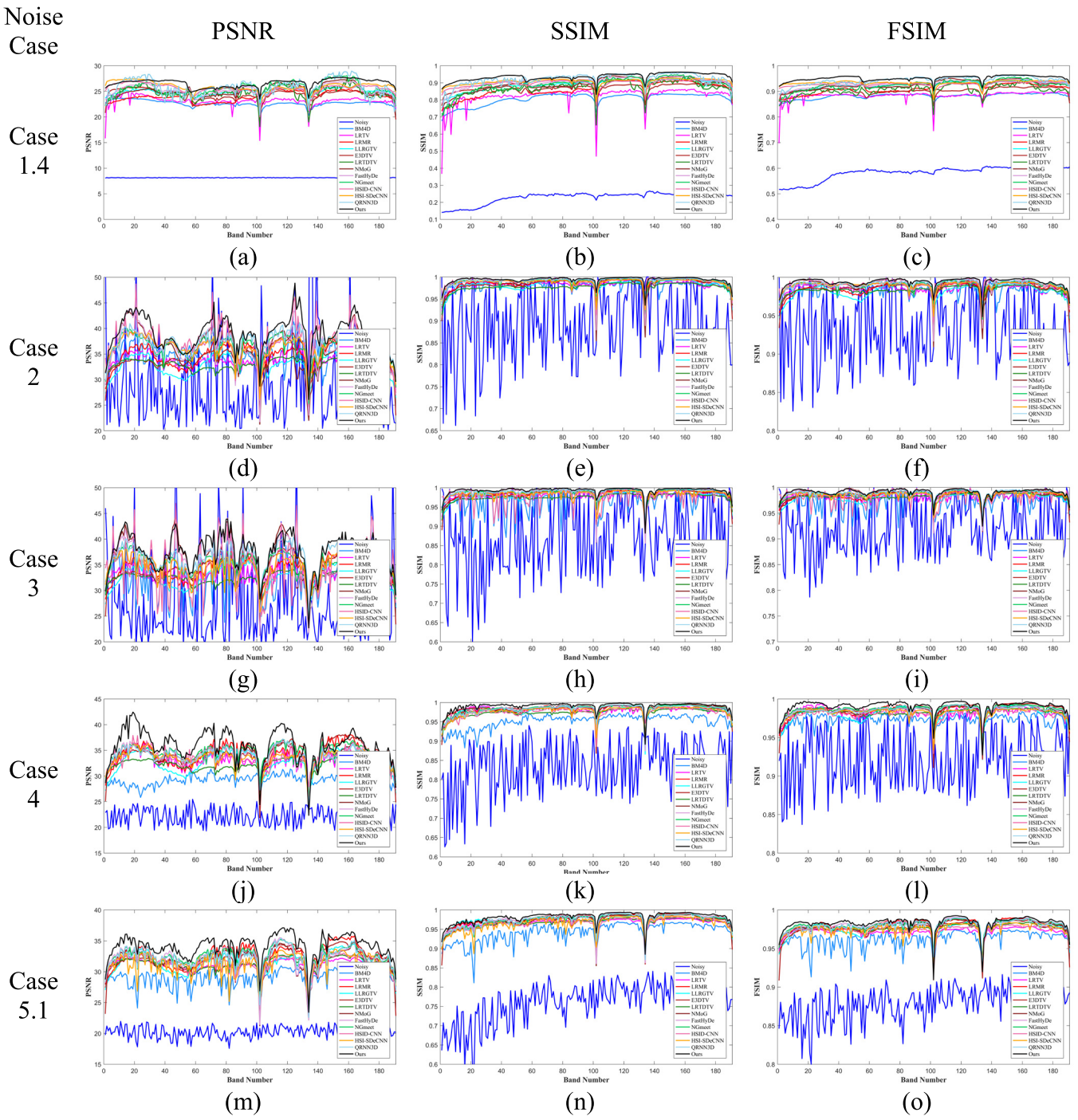

Fig. 11. (a), (d), (g), (j), and (m) PSNR; (b), (e), (h), (k), and (n) SSIM; and (c), (f), (i), (l), and (o) FSIM values of comparison methods in each band of the WDC testing dataset in Case 1.4, Case 2, Case 3, Case 4, and Case 5.1.

display the band 107 in grayscale, which is a band heavily exposed to corruptions with full band coverage. It can be observed that BM4D, LRTV, LRMR, LRTDTV, and NGmeet methods still retain highly distinct residual stripes after denoising compared with the LLRGTV method, which removes more stripes. E3DTV, HSID-CNN, and HSI-SDeCNN retain a light layer of noise mask in the image after denoising, thus making the feature information less clear. NMoG, FastHyDe, QRNN3D, and our method can recover the original feature information after removing the noise.

3) EO-1 Hyperion Dataset: The first and last few bands of the Hyperion EO-1 are heavily contaminated mainly by Gaussian and stripe noise. We display the restoration results of band 166, which is the last band, as shown in Fig. 15. The visualization in Fig. 15(n) clearly demonstrates that the proposed FATR method achieves the best performance. BM4D, LRTV, and LRMR fail completely to remove most of the stripe noise. LRTDTV, NGmeet, HSI-SDeCNN, and QRNN3D can only remove part of the stripes, as they fail to remove most of stripes. E3DTV can remove most of the stripes

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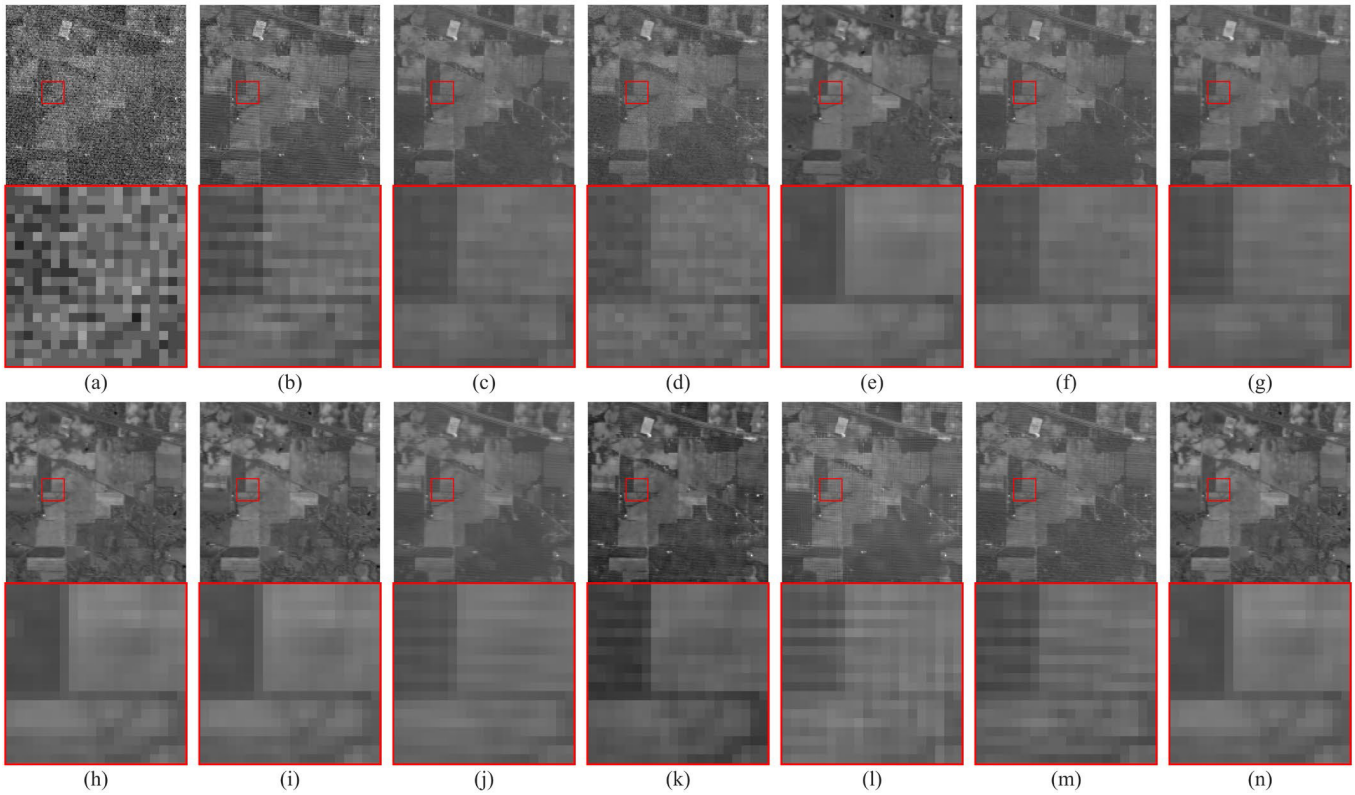

Fig. 12. Denoising results for the AVIRIS Indian Pines Site3 image in the real data experiment, grayscale display with band 1: (a) original image; (b) BM4D; (c) LRTV; (d) LRMR; (e) LLRGTV; (f) E3DTV; (g) LRTDTV; (h) NMoG; (i) FastHyDe; (j) NGmeet; (k) HSID-CNN; (l) HSI-SDeCNN; (m) QRNN3D; and (n) FATR.

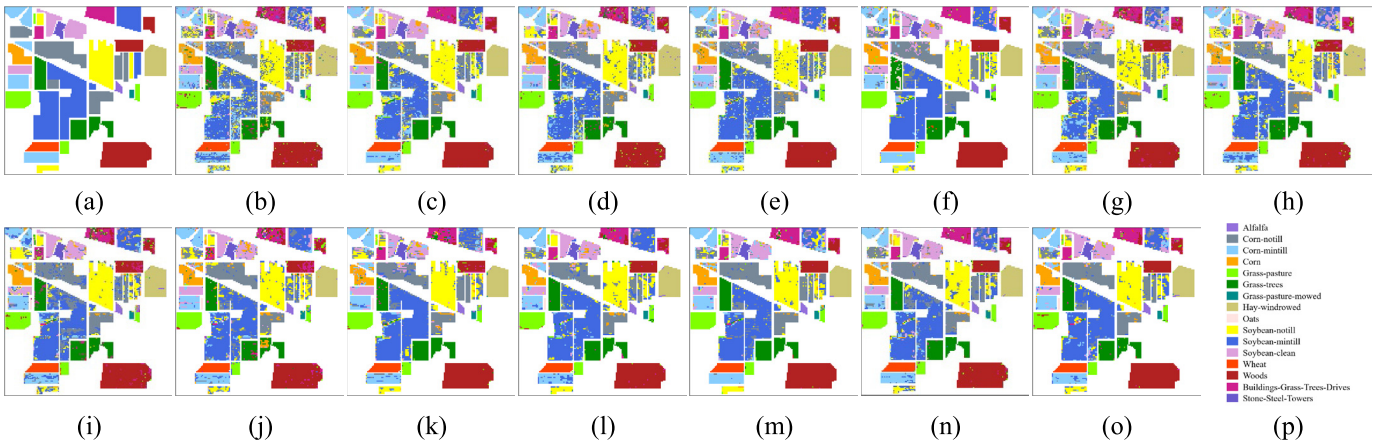

Fig. 13. Classification results for the Indian Pines Site3 image: (a) ground truth; (b) original; (c) BM4D; (d) LRTV; (e) LRMR; (f) LLRGTV; (g) E3DTV; (h) LRTDTV; (i) NMoG; (j) FastHyDe; (k) NGmeet; (l) HSID-CNN; (m) HSI-SDeCNN; (n) QRNN3D; (o) FATR; and (p) 16 classes.

but loses detailed information. In addition, NMoG, FastHyDe, and HSID-CNN could also recover well from the stripe noises.

In addition, the mean DN curve of the columns recovered in band 166 is shown in Fig. 16, where Fig. 16(a) shows the mean DN curve of the noisy image, which can be seen as a jagged fluctuation under the influence of the stripe noise. Compared with other methods, our proposed method can remove the jagged distortion and recover the original spectral information.

4) Zhuhai-1 Chaohu Dataset: The previous bands of Zhuhai-1 are seriously disturbed by the stripe noise and also mixed with slight Gaussian noise. The first few selected bands (R:8, G:4, B:2) where the noise interference is more serious are displayed in false color as shown in Fig. 17. From the results visualized in Fig. 17, BM4D cannot remove the mixture noise, and LRTV, LLRGTV, LRTDTV, FastHyDe, NGmeet, and HSID-CNN are unable to remove the stripe noise or have stripe-like color inconsistency after denoising. LRMR and NMoG can completely remove the stripe noise but do not remove the Gaussian noise. E3DTV has a spatial smoothing result with texture details lost. In addition, HSI-SDeCNN and QRNN3D remove stripe noise incompletely and also have color inconsistency after denoising, while our method could

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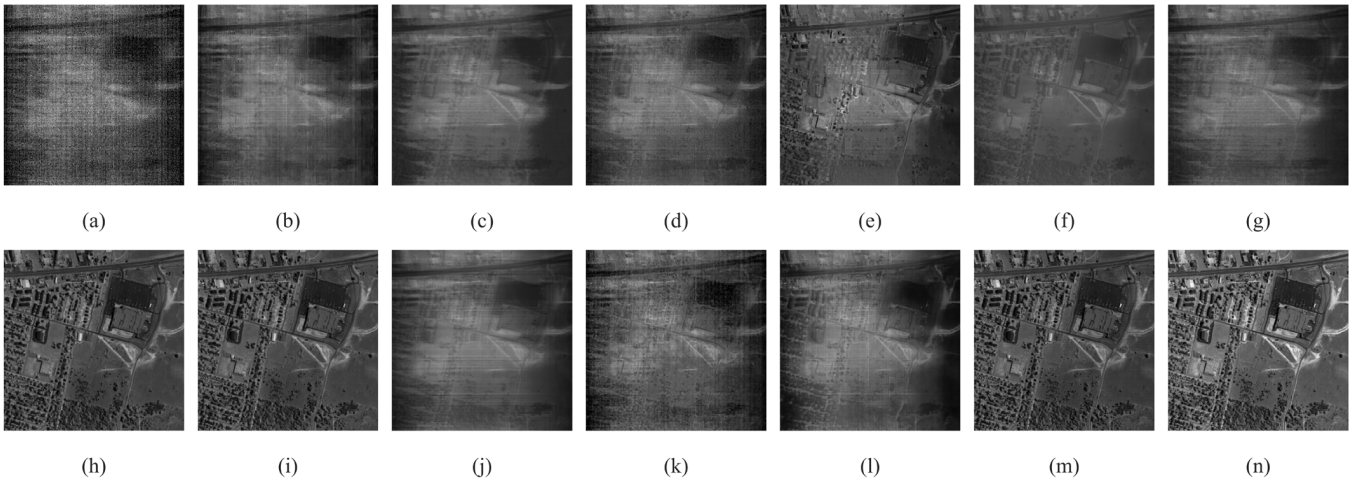

Fig. 14. Denoising results for HYDICE Urban dataset in the real data experiment, grayscale display with band 107: (a) original image; (b) BM4D; (c) LRTV; (d) LRMR; (e) LLRGTV; (f) E3DTV; (g) LRTDTV; (h) NMoG; (i) FastHyDe; (j) NGmeet; (k) HSID-CNN; (l) HSI-SDeCNN; (m) QRNN3D; and (n) FATR.

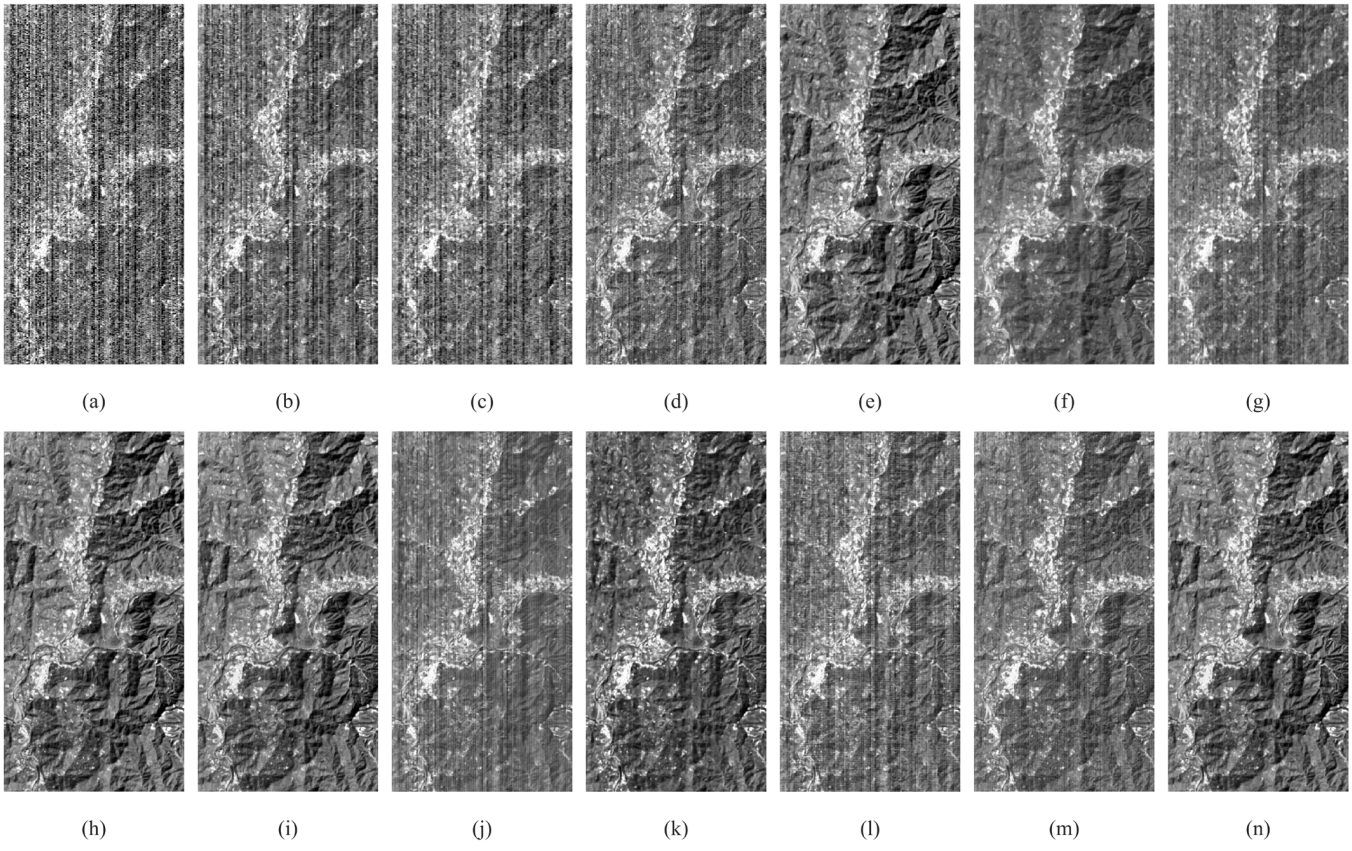

Fig. 15. Denoising results for Hyperion EO-1 dataset in the real data experiment, grayscale display with band 166: (a) original image; (b) BM4D; (c) LRTV; (d) LRMR; (e) LLRGTV; (f) E3DTV; (g) LRTDTV; (h) NMoG; (i) FastHyDe; (j) NGmeet; (k) HSID-CNN; (l) HSI-SDeCNN; (m) QRNN3D; and (n) FATR.

remove both Gaussian noise and stripe noise and preserve clear texture details.

5) GF-5 Dunhuang Dataset: The first few bands of the GF-5 data mainly suffer from heavy Gaussian noise and slight stripe noise. We display the restoration results of the first band, as shown in Fig. 18. From the visualization in Fig. 18, BM4D, LRTV, LLRGTV, and NMoG still remain with a large amount of Gaussian noises after denoising. LRTDTV, NGmeet, and HSID-CNN still retain stripe noise in the middle part of the image after removing the main Gaussian noise. E3DTV can remove a mixture of Gaussian and stripe noise at the same time but shows an over-smooth spatial result after

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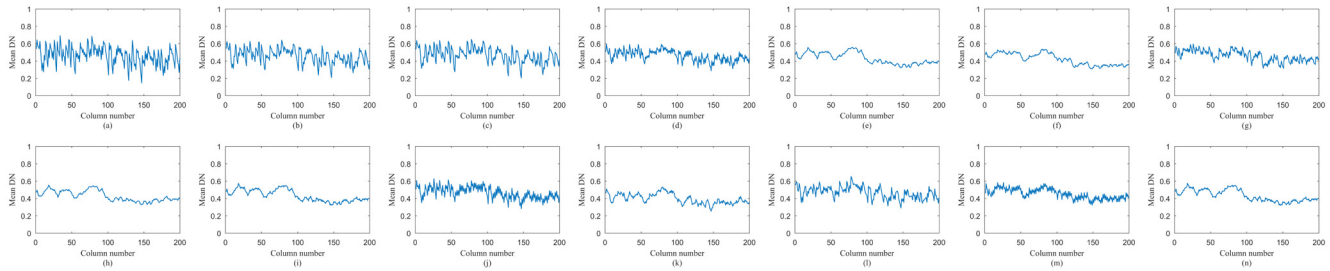

Fig. 16. Vertical mean DN profiles of band 166 from the Hyperion EO-1 dataset before and after noise removal in the real data experiment: (a) original image; (b) BM4D; (c) LRTV; (d) LRMR; (e) LLRGTV; (f) E3DTV; (g) LRTDTV; (h) NMoG; (i) FastHyDe; (j) NGmeet; (k) HSID-CNN; (l) HSI-SDeCNN; (m) QRNN3D; and (n) FATR.

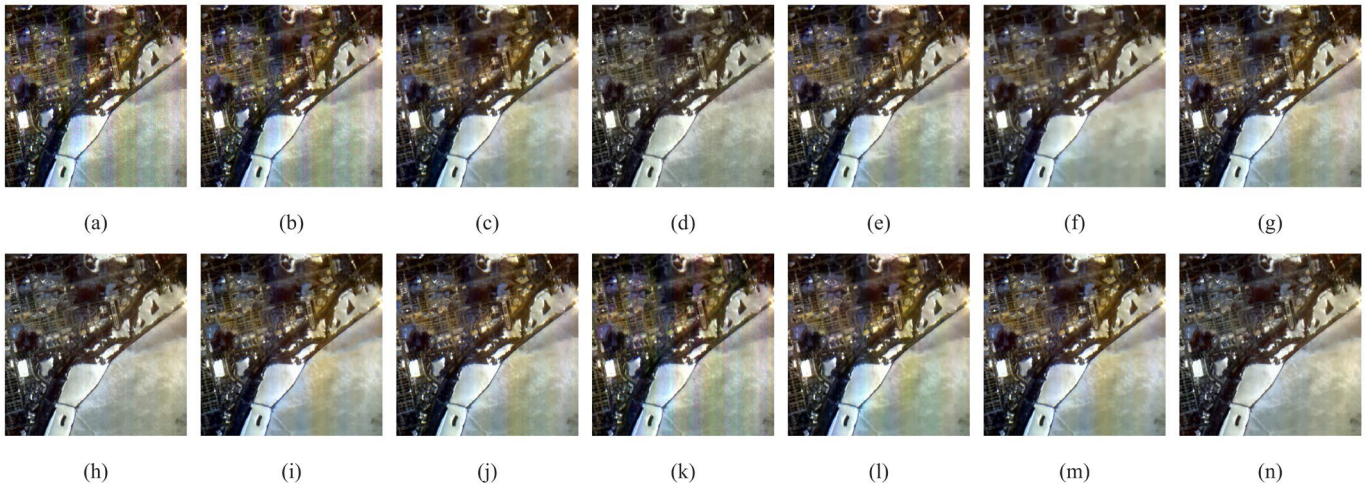

Fig. 17. Denoising results for Zhuhai-1 Chaohu dataset in the real data experiment, false color image with bands (8, 4, 2): (a) original image; (b) BM4D; (c) LRTV; (d) LRMR; (e) LLRGTV; (f) E3DTV; (g) LRTDTV; (h) NMoG; (i) FastHyDe; (j) NGmeet; (k) HSID-CNN; (l) HSI-SDeCNN; (m) QRNN3D; and (n) FATR.

TABLE II

CLASSIFICATION ACCURACY RESULTS FOR INDIAN PINES SITE3 IMAGE IN THE REAL DATA EXPERIMENT USING SVM METHOD

	Noisy	BM4D	LRTV	LRMR	LLRGTV	E3DTV	LRTDTV	NMoG	FastHyDe	NGmeet	HSID-CNN	HSI-SDeCNN	QRNN3D	FATR
OA	69.28%	81.56%	76.37%	81.30%	83.85%	77.53%	77.81%	77.92%	79.05%	82.88%	85.52%	87.17%	85.81%	87.25%
AA	0.6629	0.7972	0.7196	0.7797	0.7615	0.7230	0.7332	0.7576	0.7203	0.7804	0.8275	0.8225	0.8261	0.8353
Kappa	0.6521	0.7911	0.7319	0.7865	0.8157	0.7455	0.7476	0.7479	0.7616	0.8045	0.8353	0.8537	0.8379	0.8544

denoising. HSI-SDeCNN has a grainy residual noise in the area of the bottom right corner of the figure after denoising. LRMR, FastHyDe, QRNN3D, and our method have a better denoising result.

E. Ablation Study

1) Different Strategy of Coupling: For ablation study, we train our model on WDC training data in Case 1.1 ($\sigma = 25$) and test it on WDC test data. We show the effect of spectral embedding, spatial-lightweight, and data augmentation on model enhancement, as shown in Table III. The baseline model uses a single-band patch embedding operation and does not include the lightweighting process in the self-attention and augmentation of training data, and the window partitioning is configured as the global window strategy. For spectral embedding with four input bands in the feature extraction part, the PSNR result is improved by approximately 0.5 dB, and the MSAD is reduced by approximately 0.7. The spatial-lightweight strategy is mainly aimed at reducing the training time. Table III also shows that it would not reduce the effect of denoising while improving the efficiency of the network. For data augmentation, we use 1% and 2% linearization, rotating, and flipping as supplemental training data to enhance the training results, and it is observed that the results are improved considerably, with PSNR result improving by about 0.9 and MSAD reducing by about 1.2.

2) Spatial-Lightweight Strategy: To verify the effectiveness of the spatial-lightweight strategy, we averaged the times of ten epochs to compare the times taken before and after using it, and the results are presented in Table IV, where FATR

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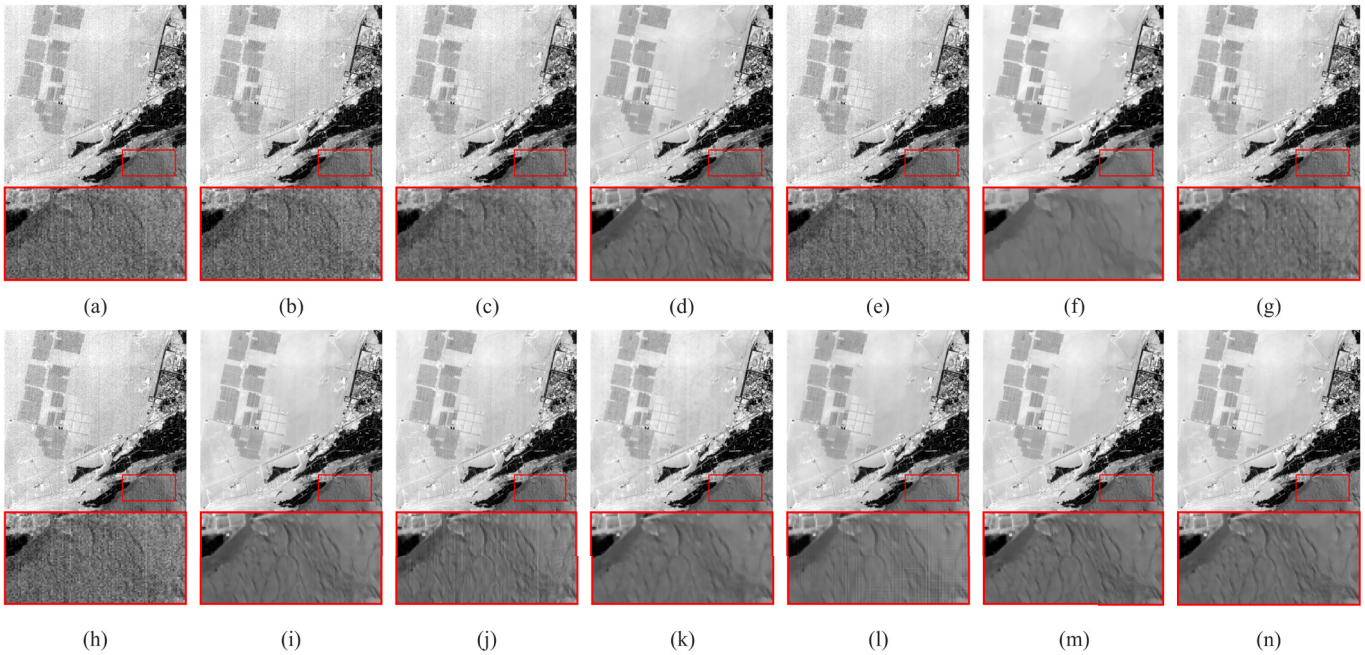

Fig. 18. Denoising results for GF-5 Dunhuang dataset in the real data experiment, grayscale display with band 1: (a) original image; (b) BM4D; (c) LRTV; (d) LRMR; (e) LLRGTV; (f) E3DTV; (g) LRTDTV; (h) NMoG; (i) FastHyDe; (j) NGmeet; (k) HSID-CNN; (l) HSI-SDeCNN; (m) QRNN3-D; and (n) FATR.

TABLE III
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT NETWORK CONSTRUCTION IN CASE 1.1 ON WDC TEST DATA

	Baseline	Baseline +SE	Baseline +SL	Baseline +DA	Regular Windows
Baseline	●	●	●	●	●
Spectral Embedding (SE)		●			●
Spatial-Lightweight(SL)			●		●
Data Augmentation(DA)				●	●
MPSNR	31.86	32.31	31.88	32.74	32.93
MSSIM	0.9721	0.9735	0.9726	0.9786	0.9798
MSAD	5.68	4.96	5.62	4.47	4.21

TABLE IV
TRAINING TIME OF EACH EPOCH IN TEN TIMES AVERAGE FOR DIFFERENT STRATEGIES WITH AND WITHOUT SPATIAL-LIGHTWEIGHT OPERATIONS IN CASE 1.1 ON WDC TEST DATA

	Baseline	Baseline+spatial-lightweight	FATR
Time of each epoch (in seconds)	92.2s	81.7s	75.4s
Iterations(10^3)	1100	750	700

denotes the final method proposed in this article. As can be seen from the results, the average time per epoch is reduced by about 10 s after employing the spatial-lightweight strategy. As for the number of iterations at convergence, after using the spatial-lightweight layer, the number of iterations is reduced by 3.5×10^5 compared to the baseline, which means that the iterations are reduced by 30%. Therefore, this strategy can reduce the training time considerably.

TABLE V
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT WINDOW PARTITIONING APPROACH IN CASE 1.1 ON WDC TEST DATA, G, L, AND S REPRESENT THE GLOBAL, LOCAL, AND SHIFTED WINDOWS LAYERS, RESPECTIVELY

	G	L+G	G+S	L+S	L+G+S
MPSNR	32.93	33.07	33.28	33.31	33.42
MSSIM	0.9798	0.9805	0.9813	0.9815	0.9823
MSAD	4.21	4.13	3.89	3.92	3.51

3) Windows Partitioning Strategy: Another important improvement is the use of the multiscale windows partitioning strategy. Compared with the baseline model, the G model is trained using global window partitioning after the spectral embedding, spatial-lightweight strategy and data augmentation are applied, and the evaluation indexes are improved by a significant degree, as shown in the last column of Table III. Since using local windows partitioning alone would result in a significant drop in efficiency, and using only shifted windows partitioning is far less effective than the G model, we use the G model as the basis for comparison in Table V.

Furthermore, when using a combination of GWTL and SWTL, the PSNR is improved by 0.21, and the MSAD is reduced by 0.32 relative to the G model, as shown in the model G + S of Table V. When using a combination of LWTL, GWTL, and SWTL through the multiscale windows

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TABLE VI
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT EMBEDDING DIMENSION IN CASE 1.1 ON WDC TEST DATA

C	64	128	192	256	320
MPSNR	29.57	32.18	33.42	32.55	31.64
MSSIM	0.9719	0.9796	0.9823	0.9807	0.9782
MSAD	4.73	3.89	3.51	3.82	4.05

partitioning scheme, the PSNR is improved by 0.14, and the MSAD is reduced by 0.38 relative to the G + S model, as shown in model L + G + S of Table V. It is obvious that when the multiscale windows partitioning scheme is implemented, the denoising results are improved in all aspects simultaneously.

4) Parameter Analysis: In the deep feature extraction module, the selection of the channel dimension of the feature map is highly important for feature extraction. Thus, we discuss the influence of the different embedding dimension C in the spatial–spectral feature extraction module. The embedding dimension C is equal to the patch size multiplied by the scale ratio R, while the patch size is equal to $4 \times 4 \times 4$. We set R to {1, 2, 3, 4, 5}, which also entails that the embedding dimension C be set to {64, 128, 192, 256, 320}. The resulting indices are presented in Table VI. As the dimension size continues to increase, denoising effects show a trend of first increasing and then decreasing. When the dimension is too small, the feature map contains only a small amount of spectral information, the feature extraction process is insufficient, and it is difficult to extract valuable features from the feature map. When the dimension is too large, the feature map contains many duplicate information, and the reconstruction results are prone to more residual noise components. Therefore, the appropriate size for the dimension is 192.

V. CONCLUSION

In this article, we have proposed a local–global feature-aware transformer network with residual connections for HSI denoising, which is the early application of the transformer network for the restoration of HSIs. Our main contributions are the novel utilization of spectral embedding, spatial-lightweight W-MSA, and multiscale windows partitioning scheme for efficient applications of the transformer in HSI denoising. Initially, the continuous bands of an HSI are converted into a feature map by spectral patch embedding. Then, the features are extracted using a combination of different window-based transformer layers. Meanwhile, we use spatial-lightweight operation during self-attention in these layers to reduce computational and memory costs. Moreover, we use residual connections to ensure training stability. Eventually, the clean bands are restored in the output projection from different branches, and then, a high-quality HSI is reconstructed. Experiments on synthetic Gaussian noise, hybrid noise, and real-world data demonstrate that our proposed method has a superior performance in both spatial and spectral restoration.

Nevertheless, the application of transformers for HSI restoration still has tremendous potential for improvement. Currently, our model is still very rudimentary, as it can only extract information from a few bands at a time due to the excessive computational and memory cost of the transformer. In the future, the efficiency improvement, light-weighting, and pruning of the transformer will continue to be a vital task.

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